

Conclusion

This analysis provides clear evidence that housing unaffordability and displacement risk in Charlotte are driven by a combination of socioeconomic, demographic, and housing-related factors. Through a structured modeling approach, both regression and classification techniques were used to better understand and predict these outcomes. The progression from a basic linear model to a more flexible Generalized Additive Model (GAM) demonstrated the importance of capturing non-linear relationships when modeling housing cost burden. The final GAM, using four key predictors, achieved strong and consistent performance, indicating that a relatively small set of variables can effectively explain much of the variation in housing burden.

Similarly, the classification analysis identified the XGBoost model as the most effective tool for predicting displacement risk. Its ability to maintain high recall ensures that the majority of at-risk neighborhoods are correctly identified, which is especially important in real-world decision-making contexts where missing vulnerable populations can have serious consequences.

Beyond predictive performance, this project highlights a deeper structural issue: the lasting impact of historical housing discrimination. The observed relationship between past redlining practices and present-day housing outcomes suggests that current disparities are not random, but rather rooted in long-standing systemic inequities. Even without explicitly including historical grading as a feature, both models reflected these patterns, reinforcing the idea that historical disadvantage continues to shape modern socioeconomic conditions.

Overall, this project demonstrates how data-driven modeling can be used not only to predict outcomes but also to uncover meaningful insights about societal challenges. While the models perform well, limitations such as sample size, cross-sectional data, and the presence of

outliers suggest that future work could benefit from incorporating longitudinal data and additional contextual variables. Nonetheless, the findings provide a strong foundation for understanding housing instability and can support policymakers, urban planners, and community organizations in making more informed, equitable decisions.